



大作业 3.

1. $\vec{B} = 0, \hat{H} = v_F \hat{\sigma} \cdot \hat{p}$

(1). $\hat{\sigma}$ 作用于自旋空间, $\hat{p}$ 作用于位置空间, 则 $[\hat{\sigma}, \hat{p}] = 0$

(2). $[p_\alpha, p_\beta] = 0, [p_\alpha, \vec{p}] = 0, [\vec{p}, \vec{p}] = 0$

$\Rightarrow$ (3). $[\hat{H}, \hat{p}] = 0$. 即 $\hat{H}, \hat{p}$ 存在共同本征态 $\psi(\vec{x}, s) = e^{i\vec{p} \cdot \vec{x}/\hbar} \chi(s)$.

对于自旋空间, 以 $|\uparrow\rangle, |\downarrow\rangle$ 为基, $\chi(s) = \psi_1 |\uparrow\rangle + \psi_2 |\downarrow\rangle = \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix}$.

(4) $\hat{H}\psi = E\psi$

(i) 由于 $\sigma_\alpha \sigma_\beta = \delta_{\alpha\beta} I + i\epsilon_{\alpha\beta\gamma} \sigma_\gamma$

则 $(\hat{\sigma} \cdot \vec{p})^2 = \sum_{\alpha\beta} \sigma_\alpha \sigma_\beta p_\alpha p_\beta = \sum_{\alpha\beta} (\delta_{\alpha\beta} I + i\epsilon_{\alpha\beta\gamma} \sigma_\gamma) p_\alpha p_\beta = \sum_{\alpha} p_\alpha^2 I = |\vec{p}|^2 I$

故 $\hat{H}$ 的本征值是 $\pm |\vec{p}| v_F$

(ii) 考虑算符 $\hat{p}$ 用过程

$(\hat{\sigma}_x \hat{p}_x + \hat{\sigma}_y \hat{p}_y + \hat{\sigma}_z \hat{p}_z) e^{i\vec{p} \cdot \vec{x}/\hbar} \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} = (\hat{\sigma}_x p_x + \hat{\sigma}_y p_y + \hat{\sigma}_z p_z) e^{i\vec{p} \cdot \vec{x}/\hbar} \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix}$

欲确定 $\begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix}$, 即求解矩阵 $\begin{bmatrix} \hat{\sigma}_x p_x + \hat{\sigma}_y p_y \\ \hat{\sigma}_x p_x + \hat{\sigma}_y p_y + \hat{\sigma}_z p_z \end{bmatrix}$ 的本征向量. 且由(i)知, 本征值为 $\pm |\vec{p}|$.

2. $\vec{B} \neq 0, \hat{H} = v_F \hat{\sigma} \cdot (\hat{p} + \frac{e}{c} \vec{A})$

取对称规范, $\vec{A} = (0, \lambda, 0) B, \vec{B} = (0, 0, B)$

(1) $\hat{H} = v_F \hat{\sigma} \cdot (\hat{p} + \frac{e}{c} \vec{A}) = v_F (\hat{\sigma}_x \hat{p}_x + \hat{\sigma}_y (\hat{p}_y + \frac{eB}{c} x) + \hat{\sigma}_z \hat{p}_z)$. 且 $[\hat{p}_y, \hat{H}] = 0, [\hat{p}_z, \hat{H}] = 0$

即 $\hat{H}, \hat{p}_y, \hat{p}_z$ 存在共同本征态, $\psi(x, y, z, s) = e^{ip_y y/\hbar} e^{ip_z z/\hbar} \chi(s) u(x)$





(2) 价键哈密顿量

$$V_F \left[\hat{\sigma}_x (\hat{p}_x u) + u (\hat{p}_y \hat{\sigma}_y + \hat{p}_z \hat{\sigma}_z) + \hat{\sigma}_y \frac{eB}{c} \hat{x} u \right] \chi = E u \chi$$

展开为矩阵形式. $\hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, $\hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $u \chi = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

令 $p_y = \hbar k_y$, $p_z = \hbar k_z$.

得 $V_F \begin{bmatrix} \hbar k_z & -i\hbar \frac{d}{dx} - i(\hbar k_y + \frac{eB}{c}x) \\ -i\hbar \frac{d}{dx} + i(\hbar k_y + \frac{eB}{c}x) & -\hbar k_z \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = E \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

无量纲化. $[\frac{eB}{c}x] = [\hbar k_y] = [p_y] = m_L T^{-1}$

得. $[\frac{eB}{c}] = m_L^{-1} \cdot [\hbar] = [m_L^2 T^{-1}]$. 则 特征长度 $\varphi = \sqrt{\frac{\hbar c}{eB}}$.

令 $\varphi = \frac{\hbar k_y + \frac{eB}{c}x}{\sqrt{\frac{eB}{c}}} = \hbar \left(k_y + \frac{x}{\varphi^2} \right)$ 则 $\hbar k_y + \frac{eB}{c}x = \hbar \frac{x + k_y \varphi^2}{\varphi^2} = \hbar \frac{\varphi}{\varphi^2}$
 $\frac{x}{\varphi} + k_y \varphi = \frac{x + k_y \varphi^2}{\varphi}$ (无量纲化).

$\frac{d}{dx} = \frac{d\varphi}{dx} \frac{d}{d\varphi} = \frac{1}{\varphi} \frac{d}{d\varphi}$.

得 $V_F \begin{bmatrix} \hbar k_z & -i\hbar \left(\frac{1}{\varphi} \frac{d}{d\varphi} + \frac{\varphi}{\varphi^2} \right) \\ -i\hbar \left(\frac{1}{\varphi} \frac{d}{d\varphi} - \frac{\varphi}{\varphi^2} \right) & -\hbar k_z \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = E \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

特征能量. $E_0 = \frac{\hbar v_F}{\varphi}$ $\varepsilon = \frac{E}{E_0}$ (无量纲能量)

$\frac{d}{dx} = \frac{d\varphi}{dx} \frac{d}{d\varphi}$

得 $H = V_F \hbar k_z \sigma_z + E_0 \begin{bmatrix} 0 & -i \left(\frac{d}{d\varphi} + \varphi \right) \\ -i \left(\frac{d}{d\varphi} - \varphi \right) & 0 \end{bmatrix}$

记 $a = \frac{d}{d\varphi} + \varphi$, $a^\dagger = \varphi - \frac{d}{d\varphi}$. $a^\dagger a = \left(\varphi - \frac{d}{d\varphi} \right) \left(\frac{d}{d\varphi} + \varphi \right) = \varphi^2 + \varphi \frac{d}{d\varphi} - \frac{d}{d\varphi} (\varphi) - \frac{d^2}{d\varphi^2}$
 $= \varphi^2 + \varphi \frac{d}{d\varphi} - \frac{d}{d\varphi} \varphi - 1 - \frac{d^2}{d\varphi^2} = \varphi^2 - \frac{d^2}{d\varphi^2} - 1$.

即 $a^\dagger a + 1 = \varphi^2 - \frac{d^2}{d\varphi^2}$.

$a a^\dagger = \varphi^2 - \frac{d^2}{d\varphi^2} + 1$

$a a^\dagger = \left(\varphi + \frac{d}{d\varphi} \right) \left(\varphi - \frac{d}{d\varphi} \right) = \varphi^2 - \varphi \frac{d}{d\varphi} + \frac{d}{d\varphi} (\varphi) - \frac{d^2}{d\varphi^2}$
 $= \varphi^2 - \varphi \frac{d}{d\varphi} + \frac{d}{d\varphi} \varphi + 1 - \frac{d^2}{d\varphi^2} = \varphi^2 - \frac{d^2}{d\varphi^2} + 1$

$[a, a^\dagger] = 2$.

不妨令 $a = \frac{1}{\sqrt{2}} \left(\frac{d}{d\varphi} + \varphi \right)$, $a^\dagger = \frac{1}{\sqrt{2}} \left(\varphi - \frac{d}{d\varphi} \right)$ $E_0 = \frac{\hbar v_F}{\varphi} \sqrt{2}$.

地址: 南京市仙林大道 163 号

邮政编码: 210023

则 $[a, a^\dagger] = 1$, $a^\dagger a = \frac{1}{2} \left(\varphi^2 - \frac{d^2}{d\varphi^2} \right) - \frac{1}{2}$, $a a^\dagger = \frac{1}{2} \left(\varphi^2 - \frac{d^2}{d\varphi^2} \right) + \frac{1}{2}$.





由 (1) 可知, $a, a^\dagger$ 是谐振子的升降算符. 设 $|n\rangle$ 是谐振子本征值为 $(n+\frac{1}{2})\hbar\omega$ 的本征态
(并且 $\vec{r} = x\hat{x} + y\hat{y}$, $x-x_0 = x + k_y y^2 = x + k_y \frac{\hbar c}{eB}$)

$$H = U_F \hbar k_z \sigma_z + \hbar \begin{bmatrix} 0 & -ia \\ ia^\dagger & 0 \end{bmatrix}$$

以 $y = \frac{x + k_y y^2}{y}$ 为自变量. $a|n\rangle = \sqrt{n}|n-1\rangle$, $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$.

(3). 不妨取 $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} c|n-1\rangle \\ d|n\rangle \end{bmatrix}$ 求解本征波函数.

$$H \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = U_F \hbar k_z \begin{bmatrix} c|n-1\rangle \\ -d|n\rangle \end{bmatrix} + \hbar \begin{bmatrix} -id\sqrt{n}|n-1\rangle \\ ic\sqrt{n}|n\rangle \end{bmatrix}$$

$$\begin{cases} U_F \hbar k_z c - i\hbar\omega\sqrt{n}d = E c \\ -U_F \hbar k_z d + i\hbar\omega\sqrt{n}c = E d \end{cases} \Rightarrow \frac{c^2}{d^2} = \frac{U_F \hbar k_z - E}{U_F \hbar k_z + E} \quad \text{又 } c^2 + d^2 = 1$$

$$\text{则 } c = \sqrt{\frac{U_F \hbar k_z - E}{2U_F \hbar k_z}}, \quad d = \sqrt{\frac{U_F \hbar k_z + E}{2U_F \hbar k_z}} \quad (\text{取正值, 只是归一化系数})$$

(4) 求解本征值

$$H^2 = U_F^2 \hbar^2 k_z^2 \mathbb{I} + \hbar^2 \begin{bmatrix} aa^\dagger & 0 \\ 0 & a^\dagger a \end{bmatrix}$$

$$aa^\dagger a|n\rangle = a^\dagger(\sqrt{n}|n-1\rangle) = n|n\rangle.$$

$$a^\dagger a^\dagger|n\rangle = a^\dagger(\sqrt{n}|n\rangle) = (n+1)|n\rangle.$$

$$H^2 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = U_F^2 \hbar^2 k_z^2 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \hbar^2 \begin{bmatrix} aa^\dagger & 0 \\ 0 & a^\dagger a \end{bmatrix} \begin{bmatrix} c|n-1\rangle \\ d|n\rangle \end{bmatrix}$$

$$= U_F^2 \hbar^2 k_z^2 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \hbar^2 \begin{bmatrix} nc|n-1\rangle \\ nd|n\rangle \end{bmatrix} = (U_F^2 \hbar^2 k_z^2 + \hbar^2 n) \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = E^2 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\text{则 } E_n^2 = U_F^2 \hbar^2 k_z^2 + \hbar^2 n.$$

$$E_n = \pm \sqrt{U_F^2 \hbar^2 k_z^2 + \hbar^2 n}.$$

$$\text{得: 本征值 } E_n = \pm \sqrt{U_F^2 \hbar^2 k_z^2 + \hbar^2 n} \quad \text{本征态 } e^{ik_y y} e^{ik_z z} \begin{bmatrix} \sqrt{\frac{U_F \hbar k_z - E_n}{2U_F \hbar k_z}} |n-1\rangle \\ \sqrt{\frac{U_F \hbar k_z + E_n}{2U_F \hbar k_z}} |n\rangle \end{bmatrix} \quad \begin{matrix} n=1, 2, 3, \dots \\ (n=0 \text{ 另论}). \end{matrix}$$

~~今年12月27日, 4+1+1, 不在前量前并.~~





(5) 讨论 $n=0$.

取 $\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ d|0\rangle \end{bmatrix}$ 即 $\frac{1}{\sqrt{2}}c=0$. 则 $-(t, d=1)$.

$$H \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = U_F^2 \hbar^2 k_z^2 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \omega^2 \begin{bmatrix} aa^\dagger & 0 \\ 0 & a^\dagger a \end{bmatrix} \begin{bmatrix} 0 \\ d|0\rangle \end{bmatrix} = U_F^2 \hbar^2 k_z^2 \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

故 $E = U_F^2 \hbar^2 k_z^2$ $E = \pm U_F \hbar k_z$

$$H \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = U_F \hbar k_z \begin{bmatrix} 0 \\ -d|0\rangle \end{bmatrix} + \omega \begin{bmatrix} 0 \\ 0 \end{bmatrix} = -U_F \hbar k_z \begin{bmatrix} 0 \\ +d|0\rangle \end{bmatrix} = -U_F \hbar k_z$$

本征值 $E = -U_F \hbar k_z$ 本征态 $\begin{bmatrix} 0 \\ |0\rangle \end{bmatrix}$. 这本征值是 $n \neq 0$ 的退化. 符合一般通式

(6) 与佐克朗道能级的讨论.

(i) 简并度.

与佐克朗道能级类似, 一个平移量 $x_0 = -ky \frac{\hbar^2}{eB} = \frac{p_y \hbar}{eB}$ 给出一个波函数.

箱归一化: $k_y = p_y = \frac{2\pi \hbar k_y}{L_y}$ $\frac{2\pi \hbar}{L_y} n_y$ $k_y = \frac{2\pi}{L_y} n_y$ $\Delta k_y = \frac{2\pi}{L_y} \Rightarrow \Delta x = +y \frac{2\pi}{L_y}$

简并度 $n = \frac{L_x}{\Delta x} = \frac{L_x L_y}{y^2 2\pi}$

单位 $\{x, y\}$ 矩形, $L_x L_y = 1$. 单位矩形内简并度 $n = \frac{1}{2\pi y^2} = \frac{eB}{2\pi \hbar c} = \frac{eB}{\hbar c} = \frac{B}{\phi} \quad \phi = \frac{hc}{e}$

→ 结果与佐克朗道能级相同

(ii) 本征值.

佐克朗道能级 $E_n = (n + \frac{1}{2}) \hbar \omega_c + \frac{p_z^2}{2m} = (n + \frac{1}{2}) \hbar \frac{eB}{mc} + \frac{p_z^2}{2m}$

本问题朗道能级 $E_n = \pm \sqrt{U_F^2 \hbar^2 k_z^2 + \omega^2 n} = \pm \sqrt{U_F^2 p_z^2 + 2 \frac{\hbar^2 U_F^2}{y^2} n}$

$$= \pm \sqrt{U_F^2 p_z^2 + 2 \frac{\hbar^2 U_F^2}{\hbar c} eB n} = \pm \sqrt{2n \hbar \frac{eB}{c} U_F^2 + p_z^2 U_F^2} = \pm U_F \sqrt{2n \hbar \frac{eB}{c} + p_z^2}$$

a) 能级间距不是线性变化. n 越大, ΔE_n 越大; 但是同样是 p_z 的连续函数. 即 z 方向没有能级

b) 存在负能级. 但是对于 $n=0$ 的基态, 只存在负能级

地址: 南京市仙林大道 163 号

邮政编码: 210023

(iii) 本征函数. 有自旋特征. 但是对于 $n=0$ 的基态只存在一个方向的自旋分量 $\psi_0 = \begin{bmatrix} 0 \\ |0\rangle \end{bmatrix}$.

